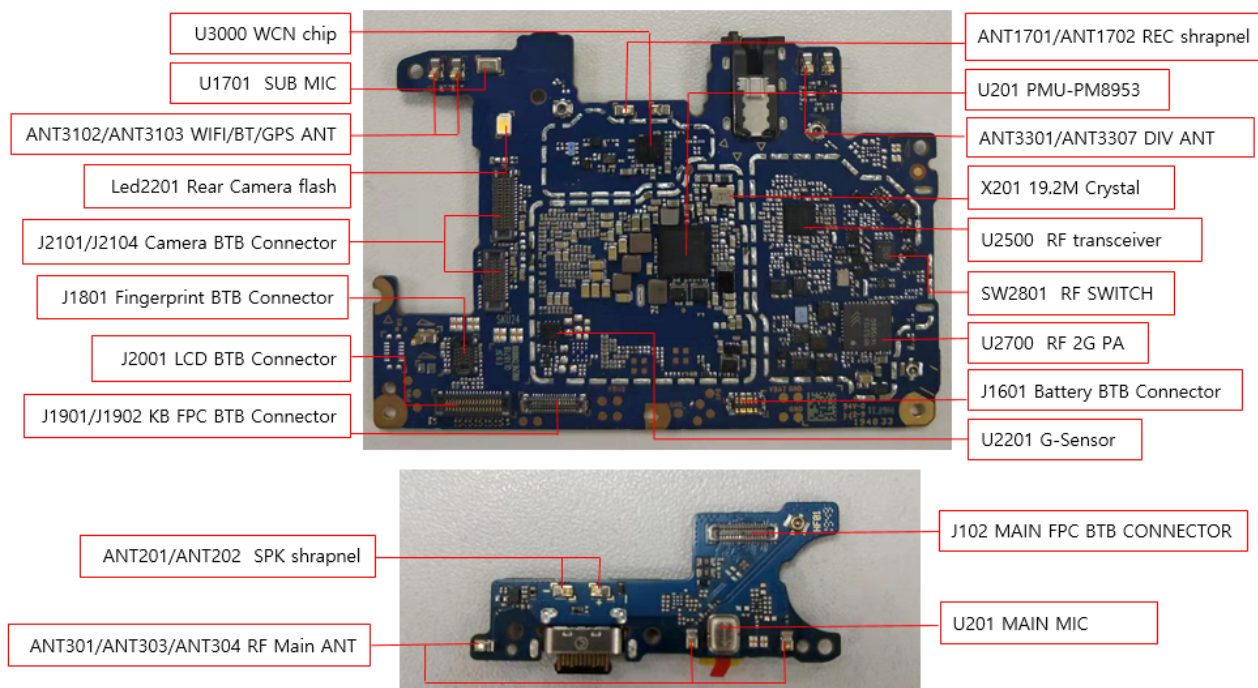


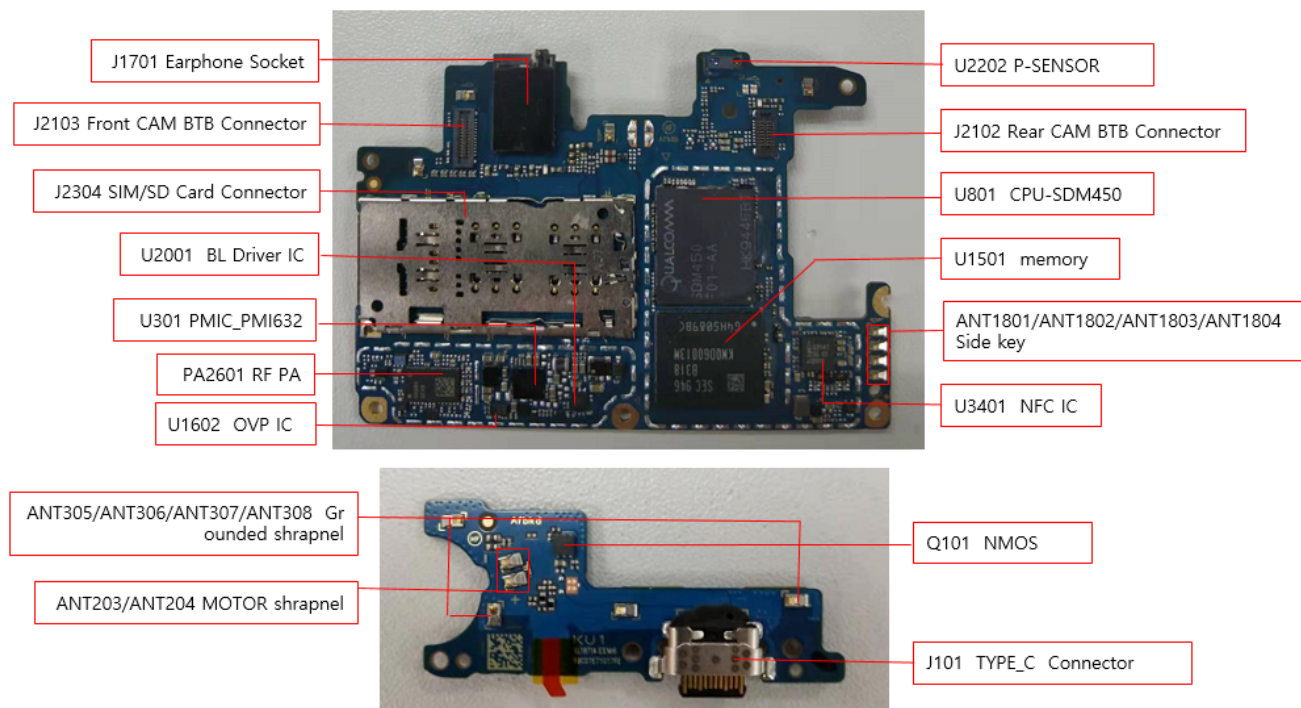
8. Level 3 Repair

8-1. Components Layout

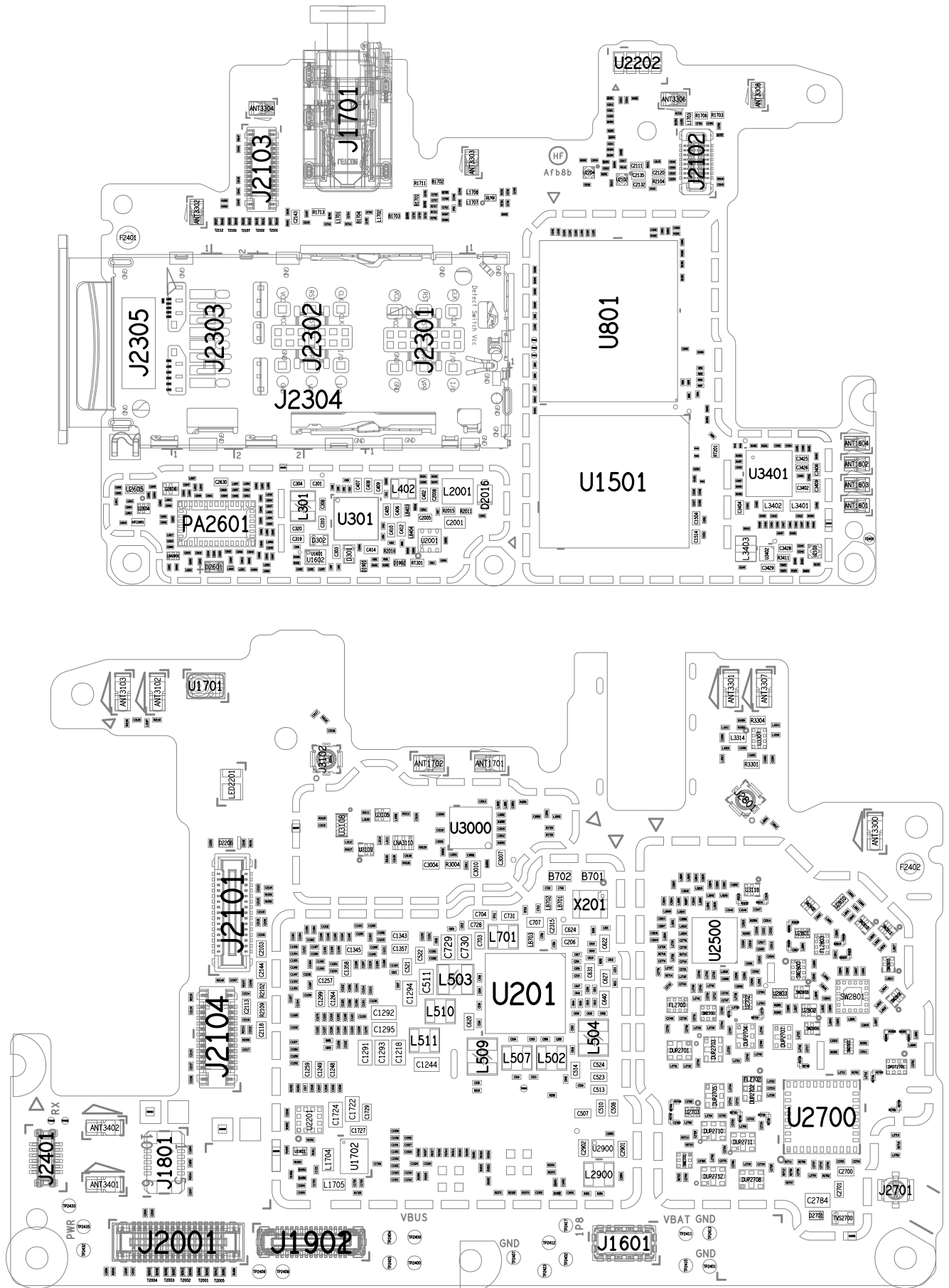
PBA (Bottom)



PBA (TOP)



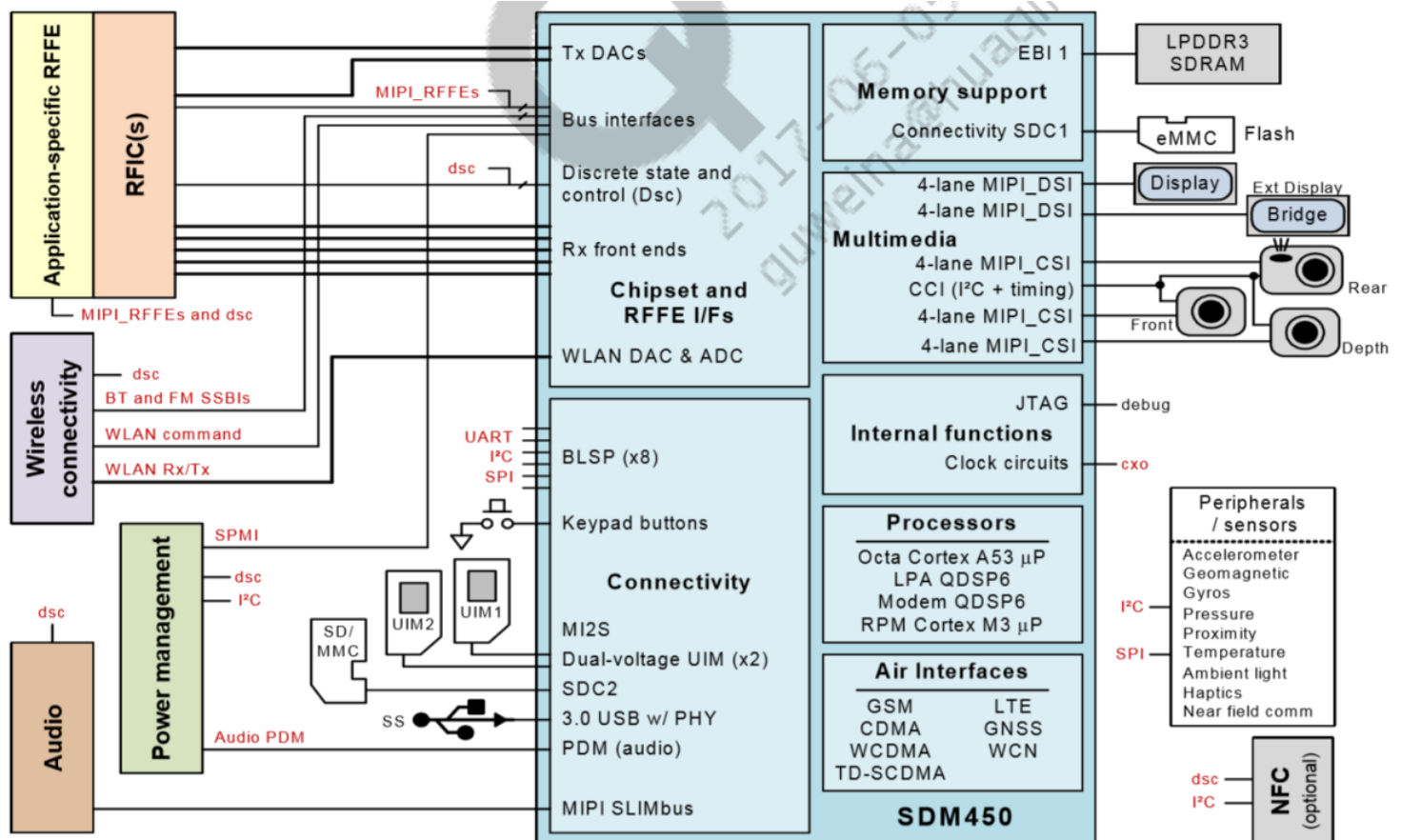
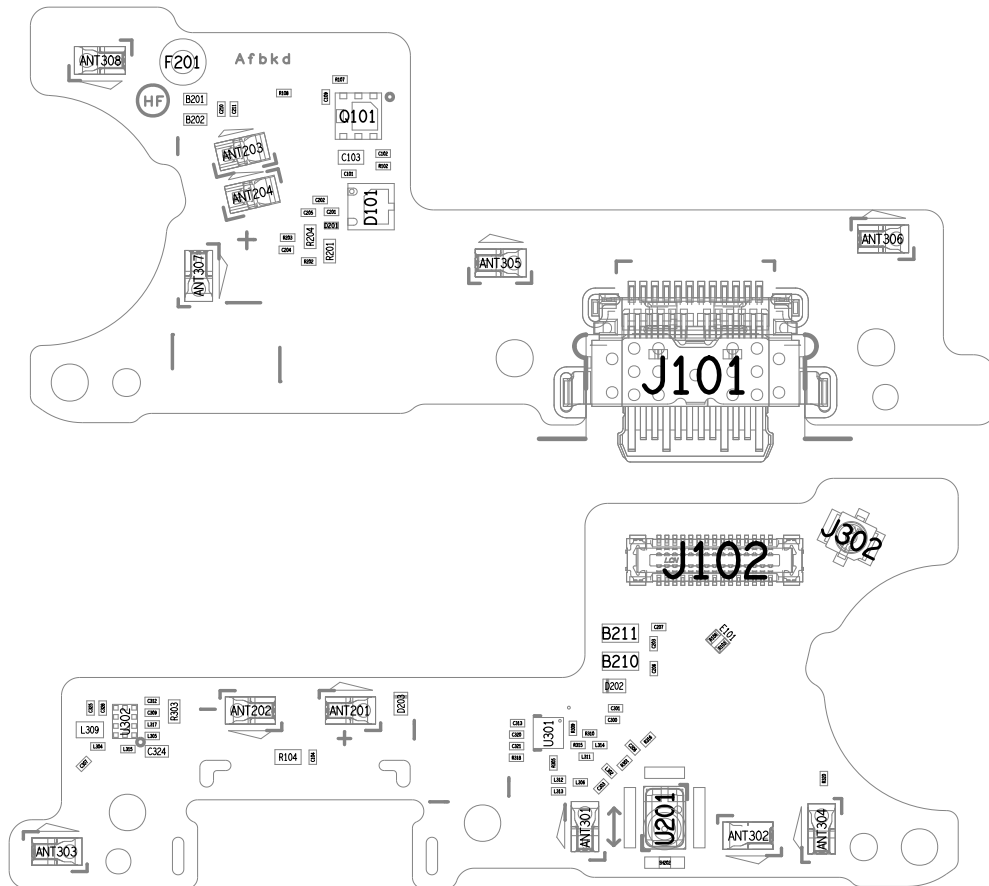
8. Level 3 Repair



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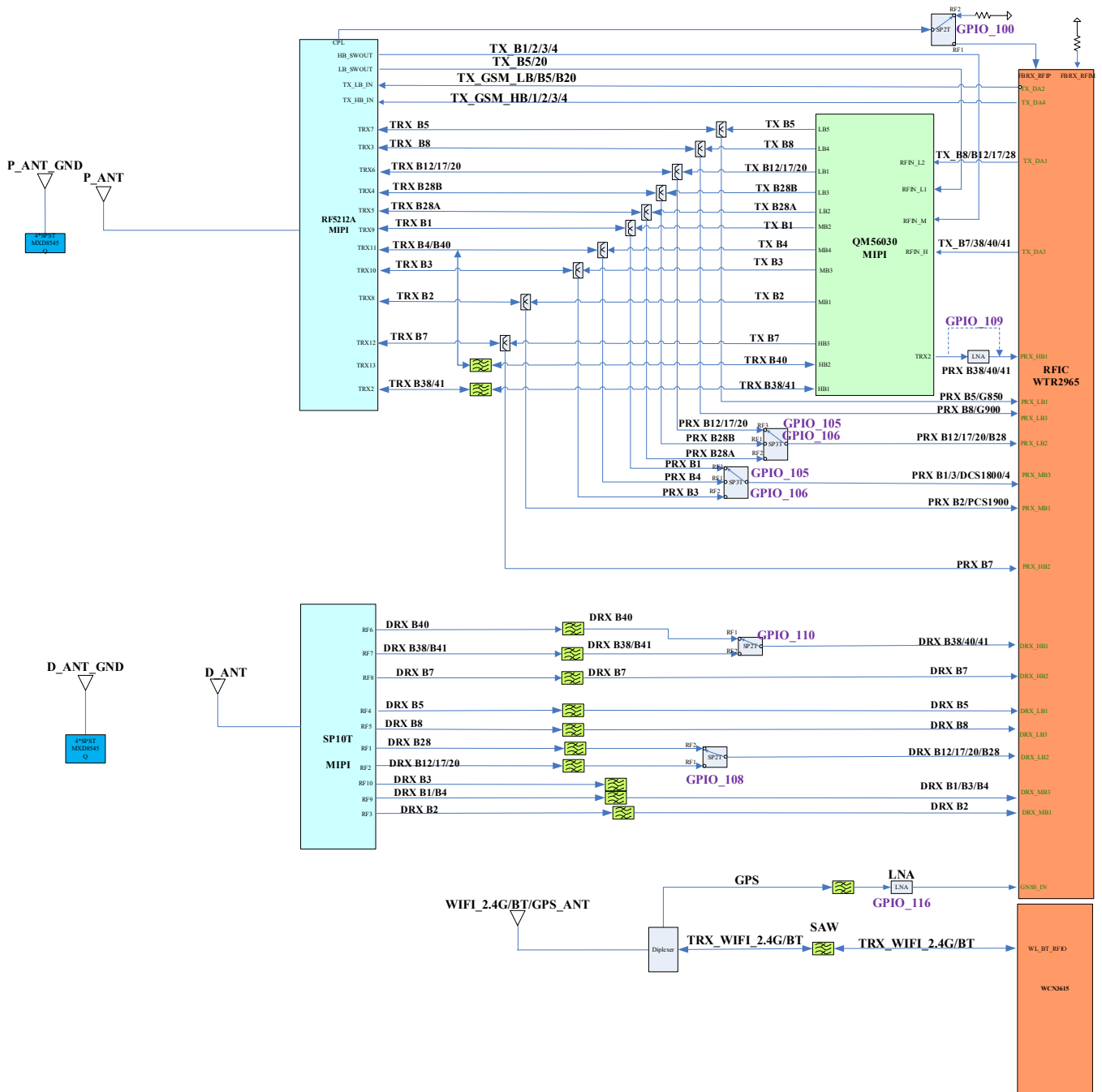
8. Level 3 Repair



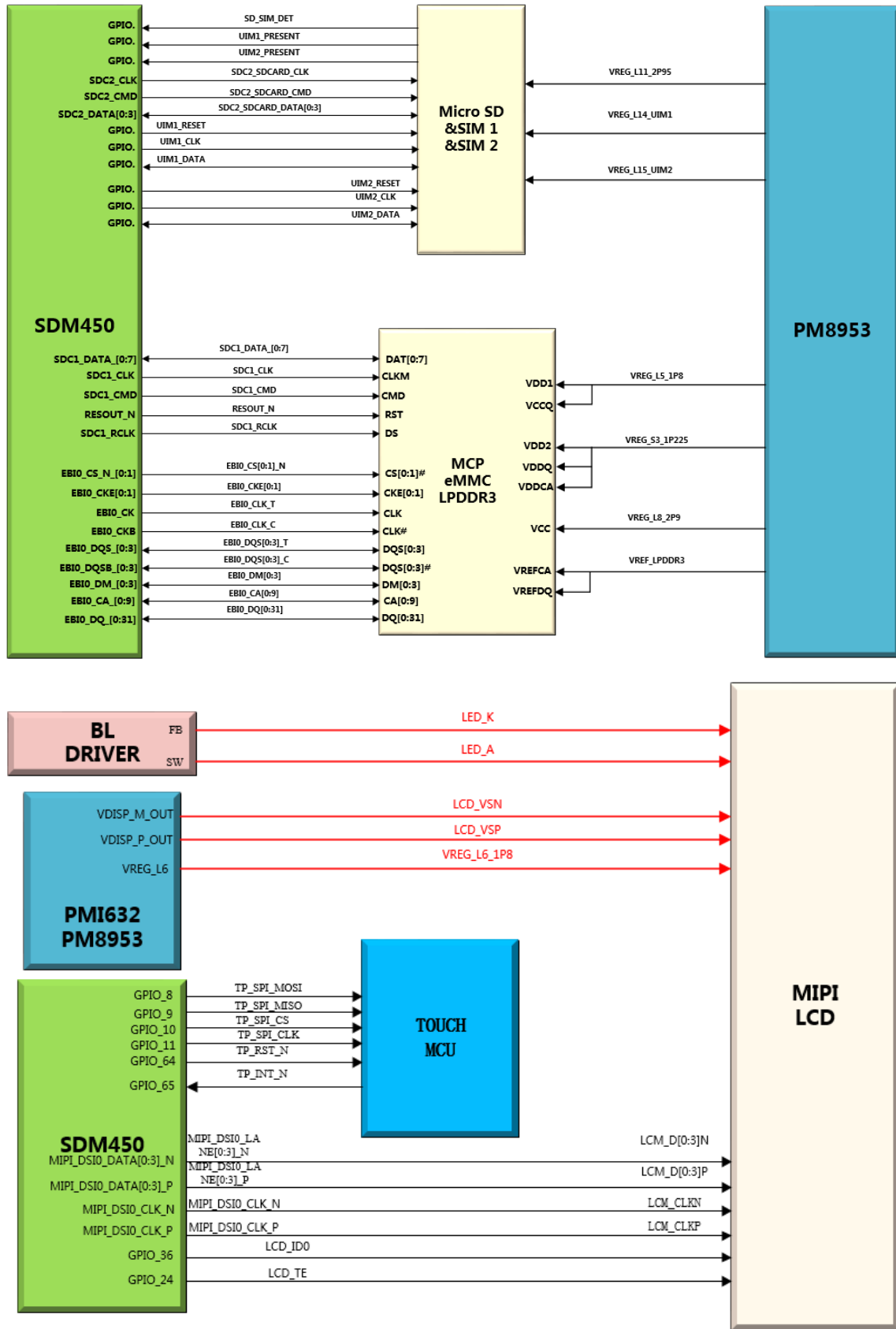
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8. Level 3 Repair



8. Level 3 Repair

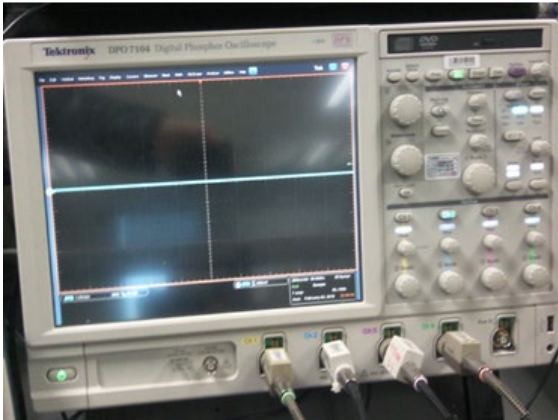


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8. Level 3 Repair

8-3. Flow chart of Troubleshooting.



Oscilloscope



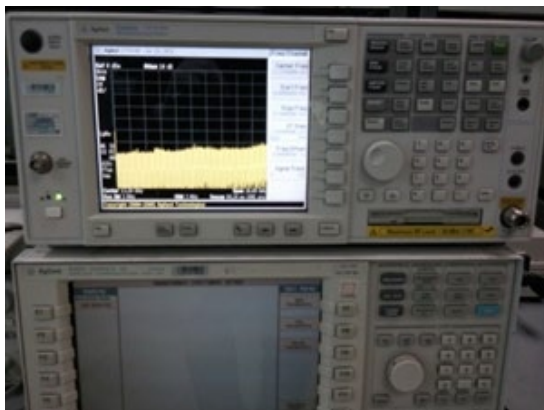
Digital Multimeter



Power Supply



+ driver, ESD Safe Tweezer



8960 & Spectrum Analyzer

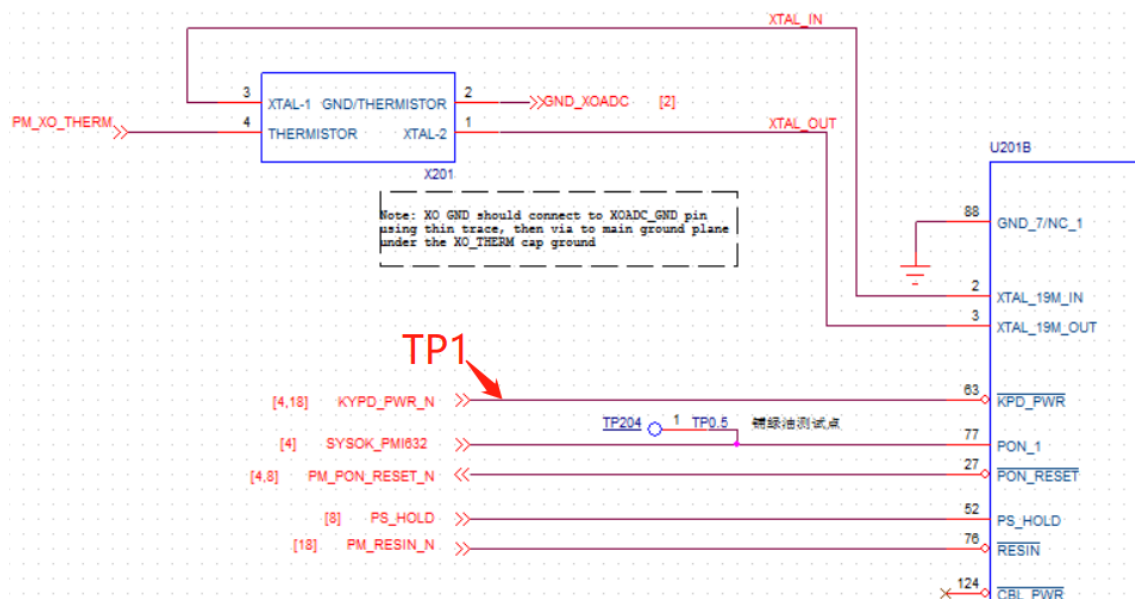
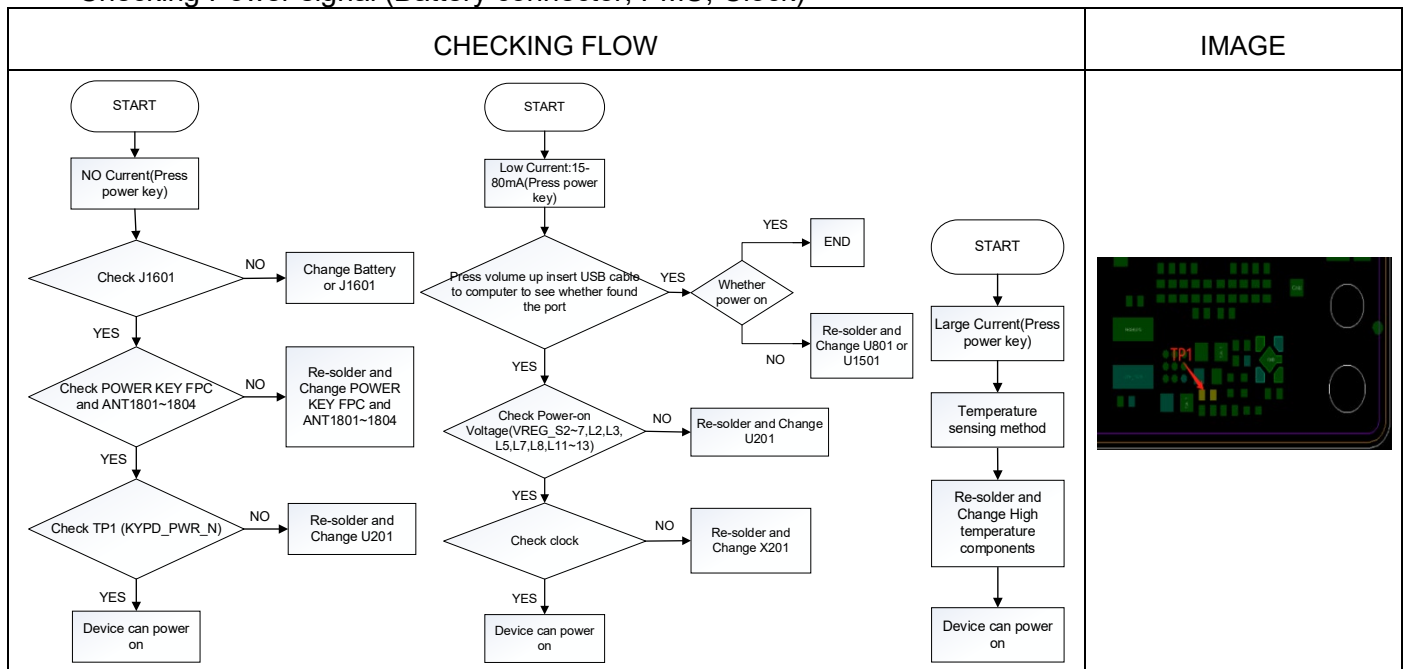


Soldering iron

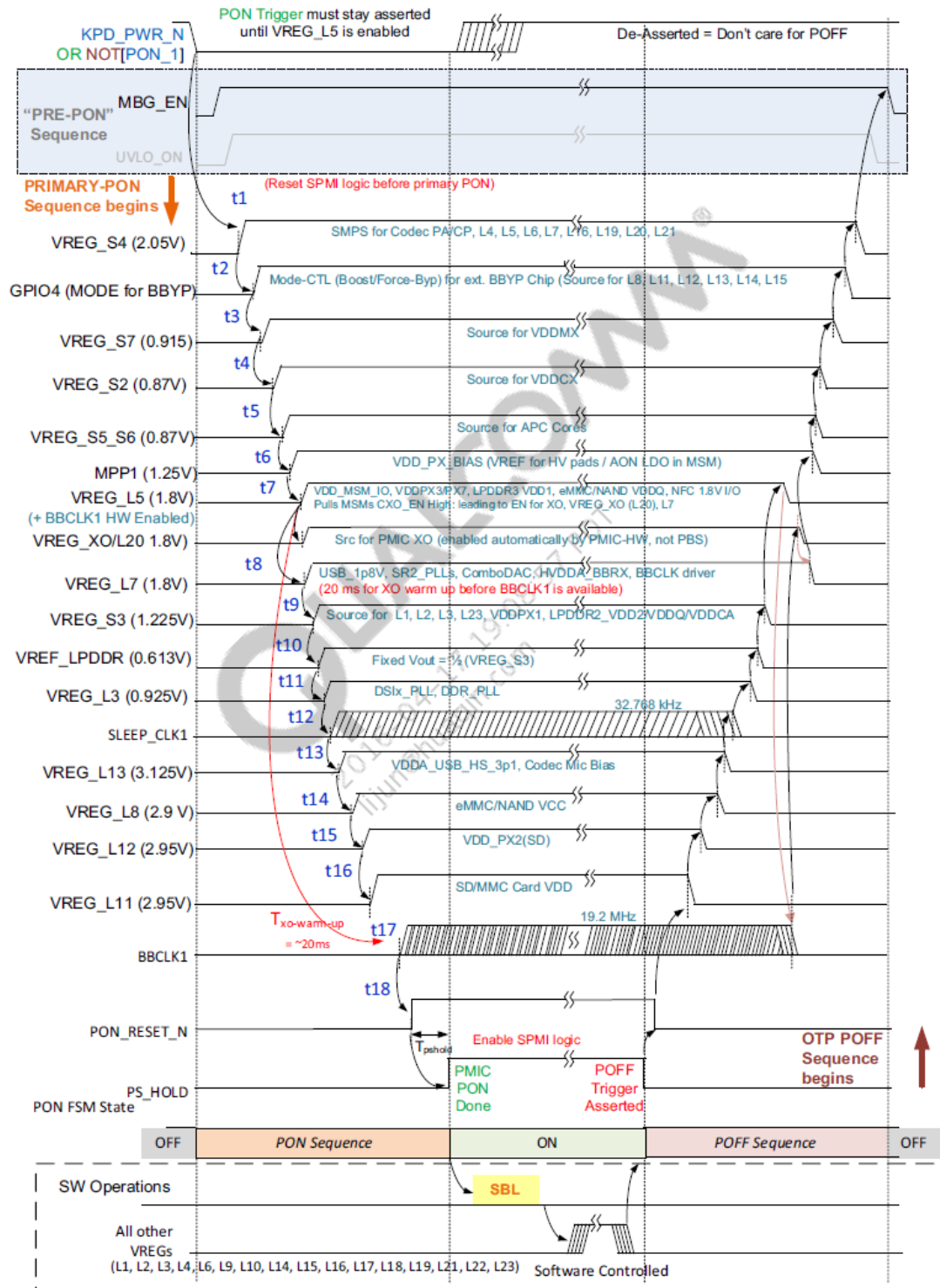
8. Level 3 Repair

8-4-1. Power On

■ Checking Power signal (Battery connector, PMU, Clock)

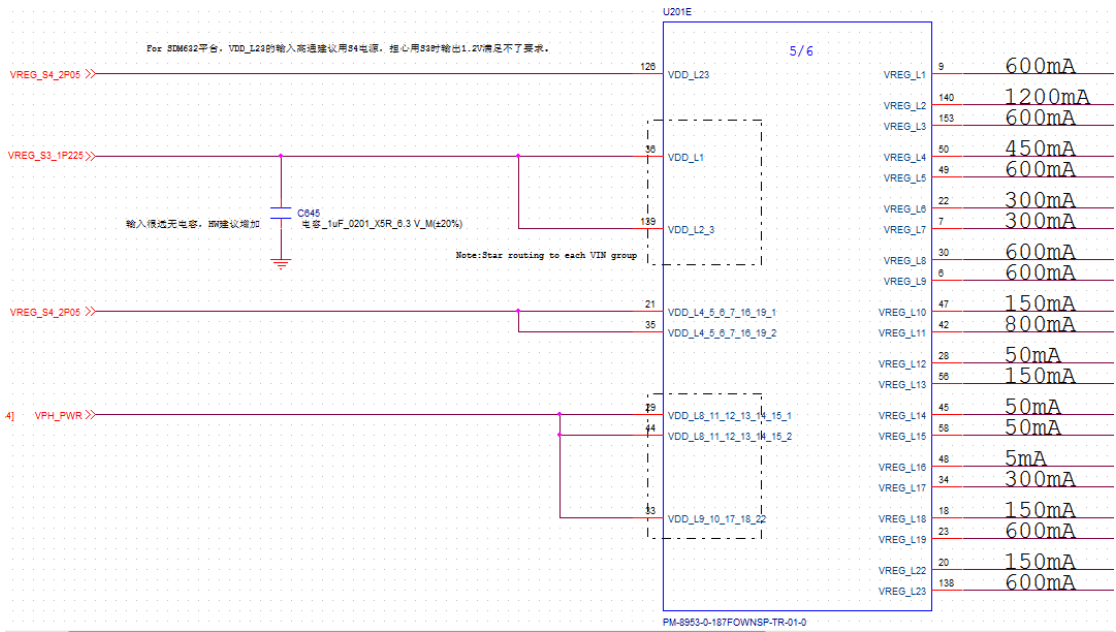


8. Level 3 Repair



Power-on Voltage (VREG_S2~7, L2, L3, L5, L7, L8, L11~13)

8. Level 3 Repair

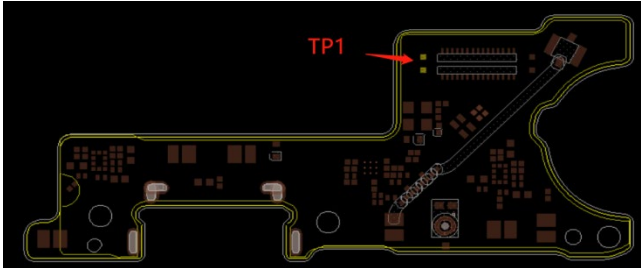
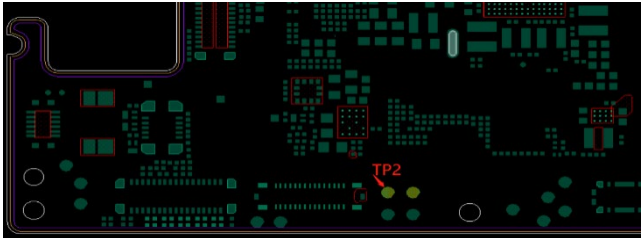
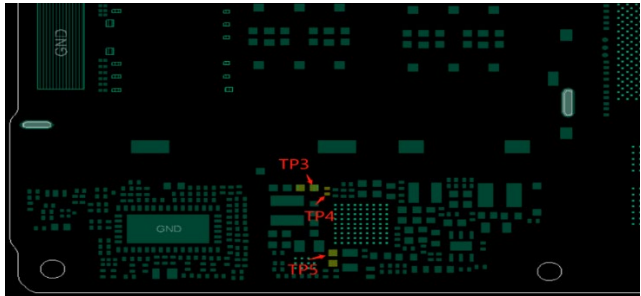


VREG_S2~7, L2, L3, L5, L7, L8, L11~13 schematic diagram

8. Level 3 Repair

8-4-2. Charging

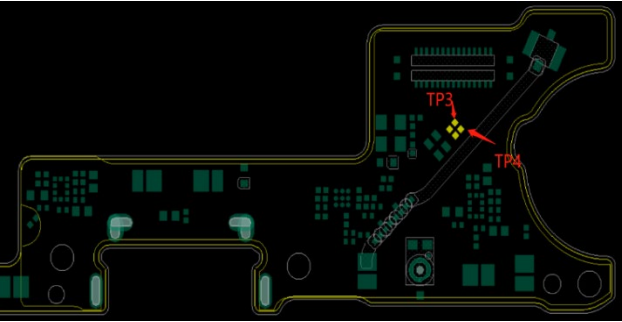
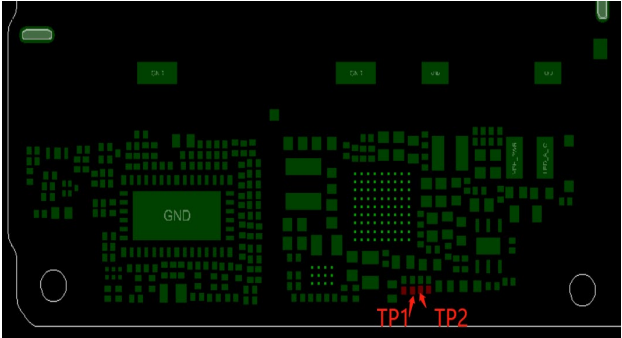
■ The charging controlled by PMU chip PMI632 (U301) and OVP chip U1602

CHECKING FLOW	IMAGE
START Check TP1 NO Re-solder and Change J102 or J101 YES Check VBUS(TP2) NO Re-solder and Change J1901 or FPC YES Check TP3 or TP4 NO Re-solder and Change U1602 YES Check Vbat(TP5) NO Re-solder and Change U301 YES Check battery YES Charge function OK	  

8. Level 3 Repair

8-4-3. USB

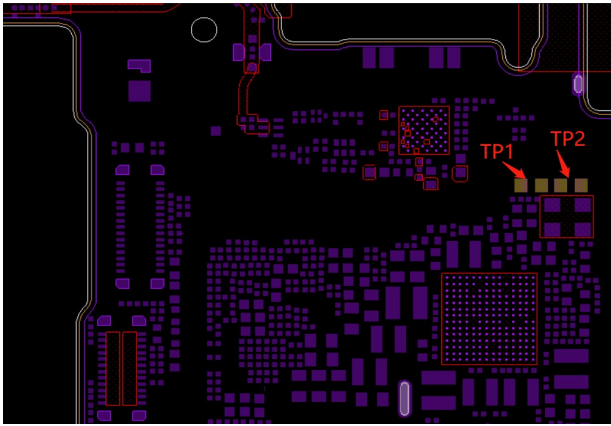
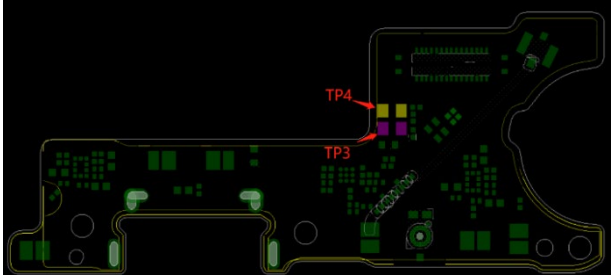
- I/O connector is used as the USB port.

CHECKING FLOW	IMAGE
START Check TP1 and TP2 NO Re-solder and Change U801 or R1901 or R1902 YES Check TP3 and TP4 NO Re-solder and Change J1901 or FPC or J102 or R103 or R106 YES Check J101 soldering NO Re-solder and Change J101 YES USB function OK	

8. Level 3 Repair

8-4-4. Audio speaker

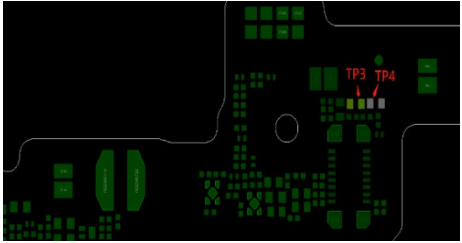
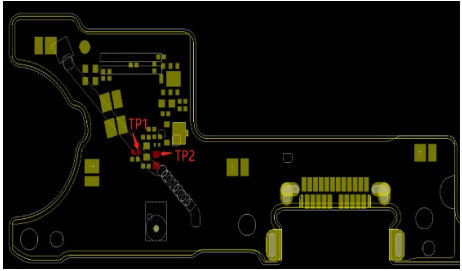
- The Speaker control signals are generated by chip PM8953 (U201), the chip and the speaker are to be checked out.

CHECKING FLOW	IMAGE
START Check TP1-TP2 NO Re-solder and Change U201 or TP1-TP2 YES Check TP3-TP4 NO Re-solder and Change J1902 or J102 or B701/702 or change FPC YES Check ANT201 and ANT202 NO Re-solder and Change ANT201/202 or B210/211 NO Change SPK spring FPC and SPK Speaker function OK	 

8. Level 3 Repair

8-4-5. Audio_ MIC

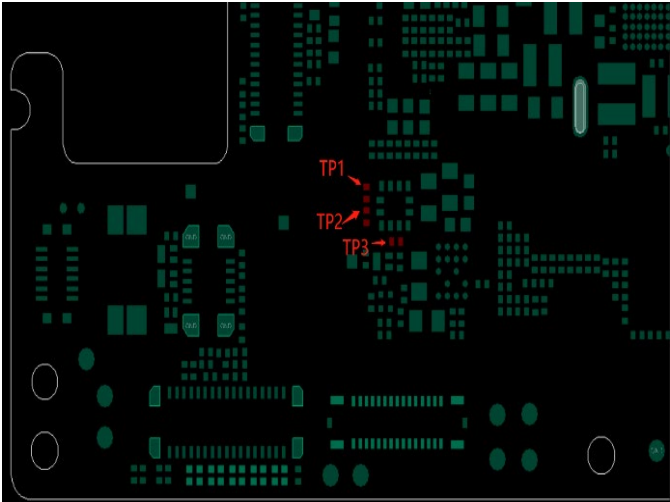
- The MIC control signals are generated by chip PM8953 (U201), the chip and the MIC (main micU201 and headset micU1701) are to be checked out.

CHECKING FLOW	IMAGE
START <pre>graph TD; START([START]) --> MainMIC[Main MIC NG]; START --> HeadsetMIC[Headset MIC NG]; MainMIC --> TP1{Check TP1}; TP1 -- NO --> ReSolderMain[Re-solder and Change U201 or J1902 or FPC or J102 or R203]; TP1 -- YES --> TP2{Check TP2}; TP2 -- NO --> ReSolderMainMIC[Re-solder and Change MIC201 or R201]; TP2 -- YES --> OKMain[MIC Function OK]; HeadsetMIC --> TP3{Check TP3}; TP3 -- NO --> ReSolderHeadset[Re-solder and Change U201 or R1706]; TP3 -- YES --> TP4{Check TP4}; TP4 -- NO --> ReSolderHeadsetMIC[Re-solder and Change U1701 or R1703]; TP4 -- YES --> OKHeadset[MIC Function OK];</pre>	

8. Level 3 Repair

8-4-6. G sensor

■ The G sensor is calibrated by using SW algorithm.

CHECKING FLOW	IMAGE
START Check VREG_L6_1P8(TP 1) NO Re-solder and Change U201 YES Check TP2-TP3 NO Re-solder and Change R1001 or R1002 YES Re-solder and Change U2201 G sensor function OK	

8. Level 3 Repair

8-4-7. Proximity and light sensor

- Proximity and Light Sensor is worked as below: Control the screen's on/off operation automatically while making phone calls, and adjust the screen brightness according to ambient light.

CHECKING FLOW	IMAGE
START Check VREG_L6_1P8(T P1&TP2) NO → Re-solder and Change R2204 or R2205 or U201 YES → Check VDD_ALPS_3V0(TP3) NO → Re-solder and Change B2203 or B2202 or U201 YES → Check TP4&TP5&TP6 NO → Re-solder and Change B2204 or B2201 or B2206 or U801 YES → Re-solder and Change U2202 ALPS function OK	

8. Level 3 Repair

8-4-11. TOUCH SCREEN

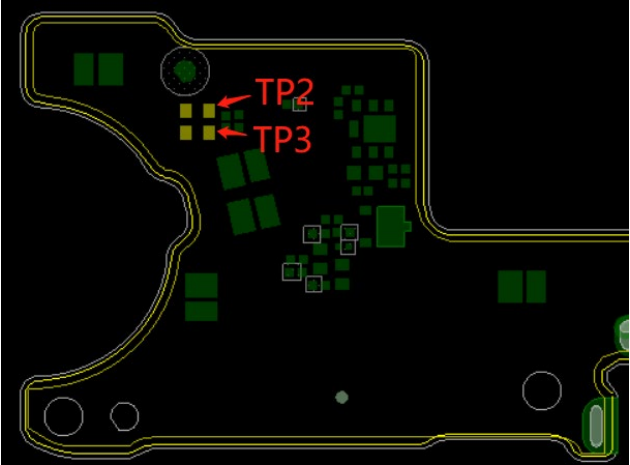
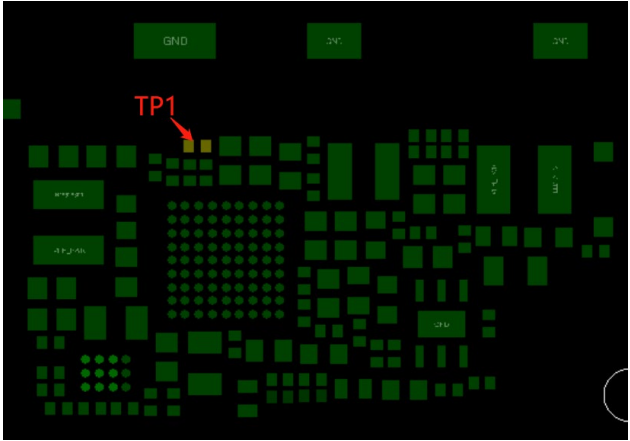
- The Touch control signals are generated by SDM450 (U801). It is assembled with LCD.

CHECKING FLOW	IMAGE
START Check TP1~TP6 NO Re-solder and Change U801 YES Check J2001 NO Re-solder and Change J2001 YES Check TP Module NO Change TP Module YES Touch function OK	

8. Level 3 Repair

8-4-12. Vibrator

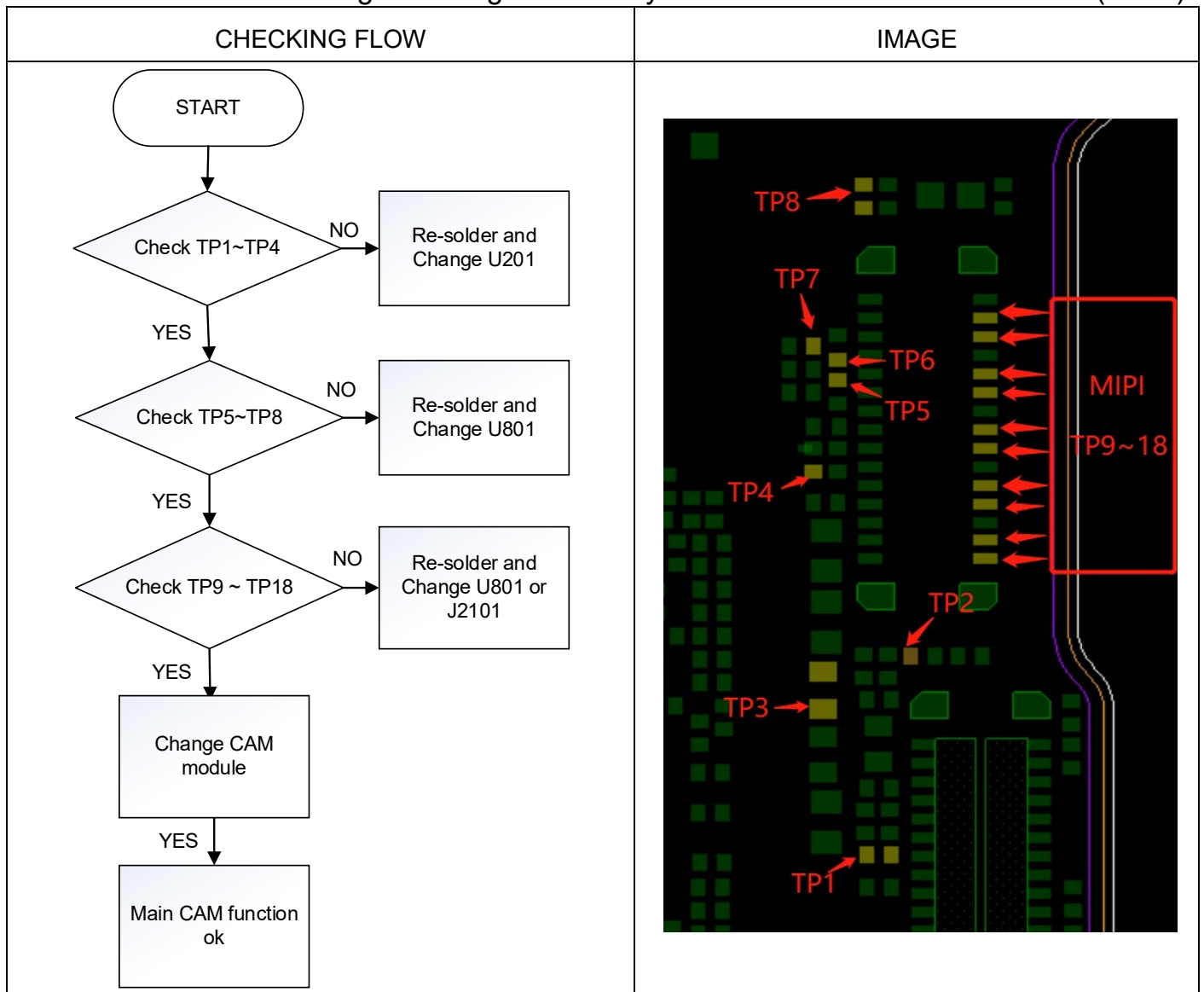
- The Vibrator control signals are generated by PMI632 (U301).

CHECKING FLOW	IMAGE
START Check TP1 NO Re-solder and Change U301 YES Check J102/ J1901/FPC NO Re-solder and Change J102/ J1901/FPC YES Check TP2,TP3 NO Re-solder and Change B201 or B202 YES Check ANT203 ANT204 and vibrator NO Re-solder and Change ANT203 ANT204 or vibrator YES Vibrator function ok	

8. Level 3 Repair

8-4-13.Main Camera

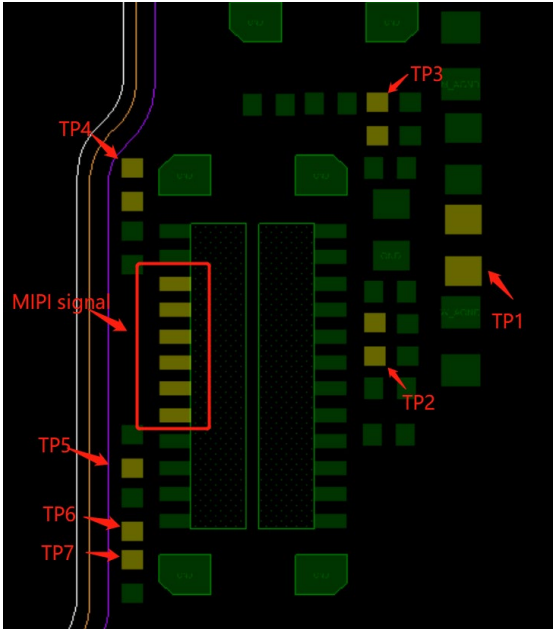
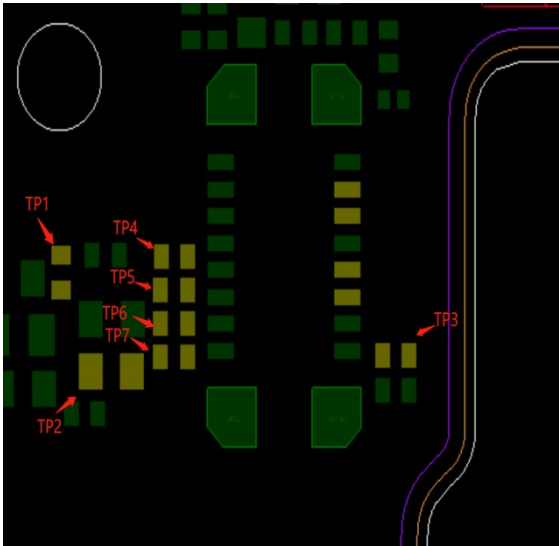
- The Camera control signals are generated by PM8953 (U201) and SDM450(U801).



8. Level 3 Repair

8-4-14. Rear auxiliary Camera

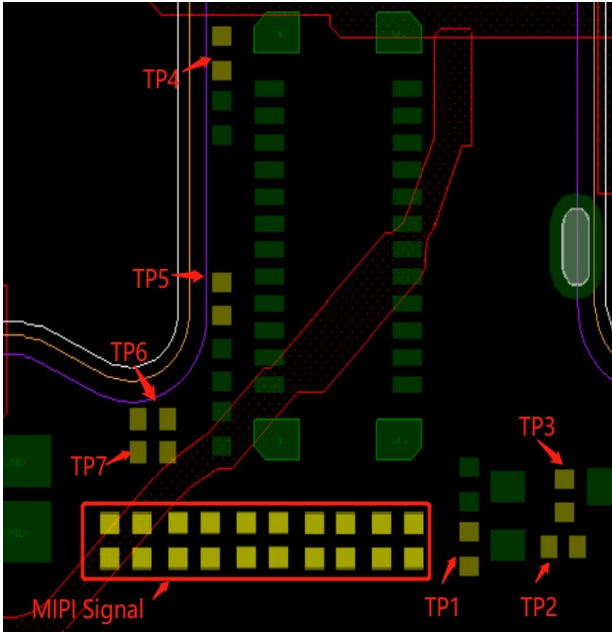
■ The Camera control signals are generated by PM8953 (U201) and SDM450(U801).

CHECKING FLOW(5M Wide CAM)	IMAGE
START Check TP1~TP3 NO Re-solder and Change U201 YES Check TP4~TP7 NO Re-solder and Change U801 YES Check MIPI signal NO Re-solder and Change U801 or J2104 YES Change CAM module YES 5M Wide CAM function ok	
CHECKING FLOW(2M Depth CAM)	IMAGE
START Check TP1~TP2 NO Re-solder and Change U201 YES Check TP3,TP5~TP7 NO Re-solder and Change U801 YES Check MIPI signal NO Re-solder and Change U801 or J2102 YES Change CAM module YES 2M Depth CAM function ok	

8. Level 3 Repair

8-4-15. Front Camera

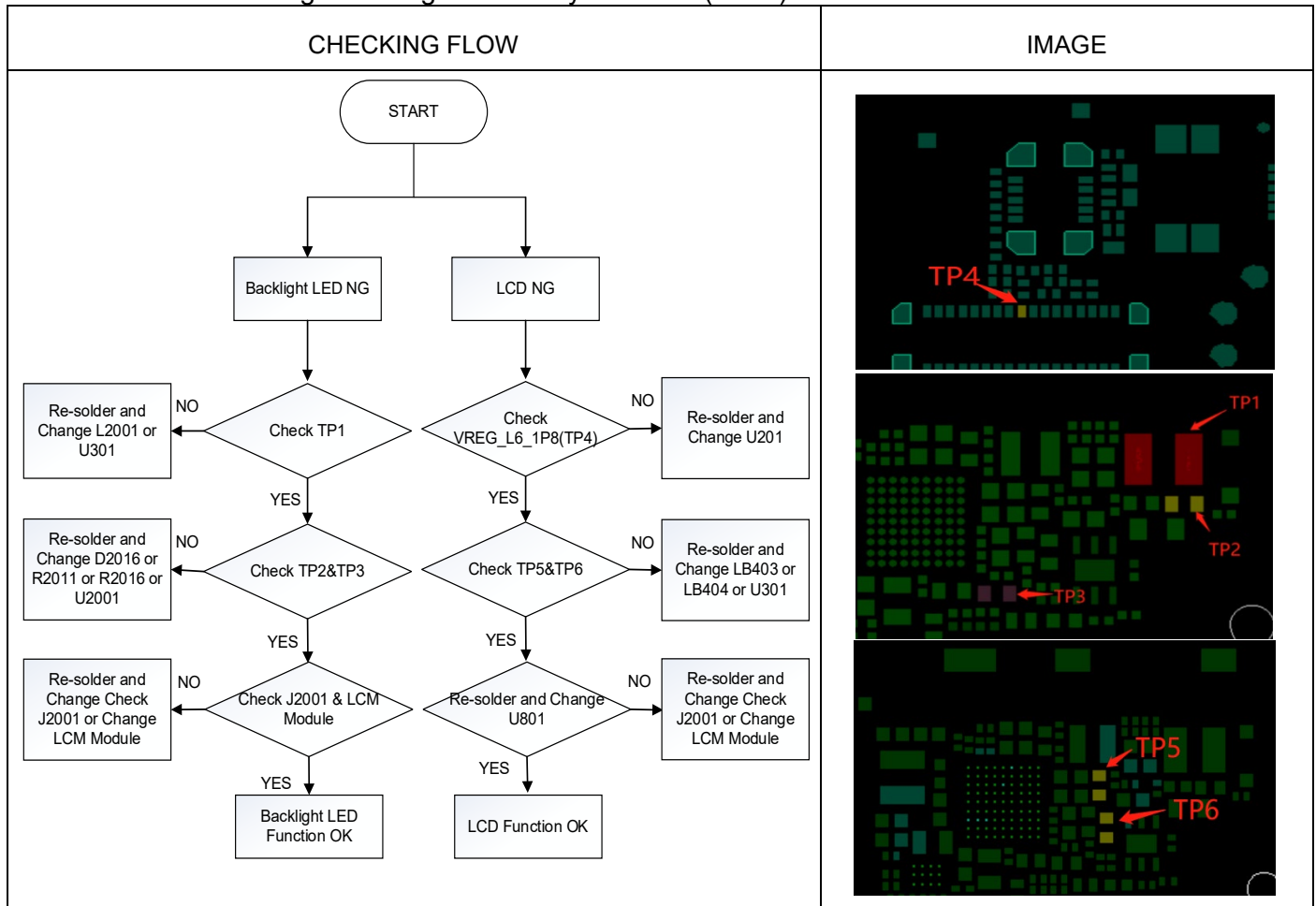
- The Camera control signals are generated by PM8953 (U201) and SDM450(U801).

CHECKING FLOW(Front CAM)	IMAGE
START Check TP1~TP3 NO → Re-solder and Change U201 YES → Check TP4~TP7 NO → Re-solder and Change U801 YES → Check MIPI signal NO → Re-solder and Change U801 or J2103 YES → Change CAM module YES → Front CAM function ok	 The image shows a microscopic view of the front camera module. A red rectangle highlights the MIPI signal line. Red arrows point to test points TP1, TP2, TP3, TP4, TP5, TP6, and TP7. The MIPI signal line is labeled 'MIPI Signal'.

8. Level 3 Repair

8-4-16. LCD

- The LCD control signals are generated by SDM450 (U801).



8-5. Service Schematics

[illegible]